

4.2 Forces: Density & Pressure

Question Paper

Course	CIEA Level Physics
Section	4. Forces, Density & Pressure
Topic	4.2 Forces: Density & Pressure
Difficulty	Medium

Time allowed: 70

Score: /58

Percentage: /100

Question 1a

A holiday company marketing 'unforgettable experiences' needs to know the maximum safe heights from which guests can jump into the sea from the end of a pier. Guests must be guaranteed they will not sink deep enough into the water to hit the bottom.

To investigate this, the designer tests how far into a measuring cylinder of water a wooden rod will sink when dropped from various heights.

The designer sets up their equipment as shown in Fig. 1.1.

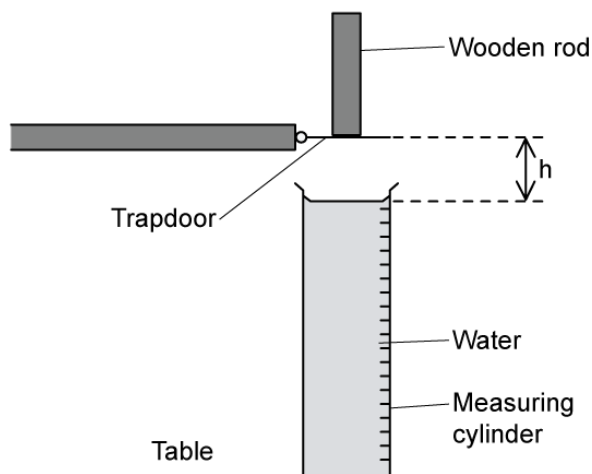


Fig. 1.1

The wooden rod has mass 1.30×10^{-2} kg, diameter 1.5×10^{-2} m and length 7.5×10^{-2} m.

(i)

Calculate the density of the wood.

[2]

(ii)

State and explain why the designer chose this type of wood to use in their investigation.

[2]

(iii)

Suggest one change to the experimental set-up which the designer needs to make to model the situation they are investigating.

[2]

[6 marks]

Question 1b

The wooden rod is held in a clamp so that it can be quickly released from rest.

Initially, it is clamped so that the base is at a height $h = 0.20$ m above the water surface.

Calculate the speed of the rod as its base reaches the water.

Ignore air resistance.

[2 marks]

Question 1c

Fig. 1.2 shows the wooden rod fully submerged under the water surface.

The rod is moving vertically downwards and has not yet come to rest.

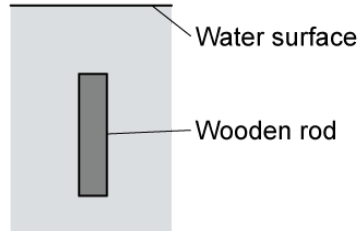


Fig. 1.2

(i)

Sketch on Fig. 1.2 the forces acting on the wooden rod.

[3]

(ii)

Describe and explain how the resultant force on the wooden cylinder varies from the moment the cylinder is fully submerged until it reaches its deepest point.

[3]

[6 marks]

Question 1d

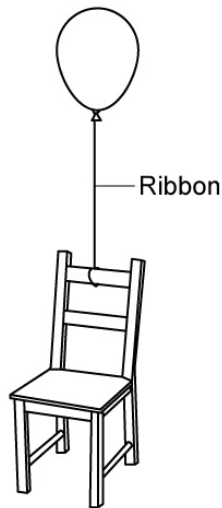
The investigation concludes that an exciting but safe experience can be had by jumping into the water which is at least six metres deep. The designer suggests a minimum age of 15 years for guests taking part in the experience.

The density of the seawater is 1.03 kg m^{-3} .

Assuming that guests will sink to no more than four metres, jumping feet first, suggest a likely range of pressure changes they will feel in their ears.

[4 marks]**Question 2a**

Party shops sell helium-filled balloons for various events. One such balloon is shown in Fig. 1.1.

**Fig. 1.1**

The ribbon is tied to a chair to stop the balloon floating away.

By modelling the balloon as a point mass, sketch a free body force diagram for the balloon and ribbon.

[2 marks]**Question 2b**

At its widest point, the balloon has a diameter 40 cm. The density of the surrounding air is 1.30 kg m^{-3} .

Calculate the upthrust on the balloon.

[3 marks]**Question 2c**

Balloons filled with air do not float.

Explain why a balloon filled with helium floats.

[4 marks]

Question 3a

The Galileo thermometer was invented in the 17th century and it is still popular today. The thermometer consists of a column of liquid containing roughly spherical, weighted, glass bulbs, as shown in Fig. 1.1.

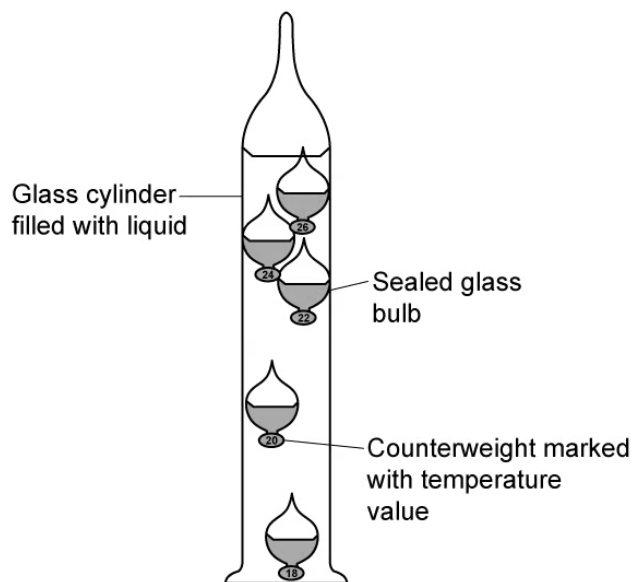


Fig. 1.1

The spheres move up or down in the column as the temperature changes.

At a certain temperature, one of the spheres begins to fall through the liquid at an increasing speed.

(i)

State the property of the liquid that causes the spheres to move up or down as the temperature changes.

[1]

(ii)

The sphere has a radius of 1.2 cm and once settled, it displaces a total mass of 7.3 g of the liquid.

Calculate the value of the property of the liquid named in (a)(i).

[3]

[4 marks]

Question 3b

Sketch arrows on Fig. 1.2 to show the relative sizes of the forces acting on the sphere as it falls.



Fig. 1.2

[4 marks]

Question 3c

Show that the upthrust on the sphere is about 0.07 N.

[2 marks]

Question 3d

The spheres in the column are identical but have different masses due to the counterweight attached to them.

State and explain where you would expect to find a sphere of weight 0.069 N.

[2 marks]

Question 4a

A submarine below the surface of the sea is stationary in a region where the density of the seawater is $1.03 \times 10^3 \text{ kg m}^{-3}$.

The submarine has a volume of $5.83 \times 10^3 \text{ m}^3$.

Calculate the upthrust exerted on the submarine by the seawater.

[2 marks]

Question 4b

Explain why the mass of the submarine must be $6.0 \times 10^6 \text{ kg}$.

[2 marks]

Question 4c

The submarine moves into a region of the sea where the water is less salty, and the density of the water reduces to $1.01 \times 10^3 \text{ kg m}^{-3}$.

Explain what would happen to the submarine as it moves into this region of lower density seawater.

[3 marks]

Question 4d

The submarine alters its weight by pumping water in or out of its internal tanks.

Determine the mass of water that the submarine should pump, in or out of its tanks, to maintain its depth below the surface of the sea.

[2 marks]

Question 5a

A hollow brass cylinder with closed ends is floating on the surface of the water.

The cylinder has a length of 4.0 cm and an external diameter of 2.1 cm as shown in Fig. 1.1.

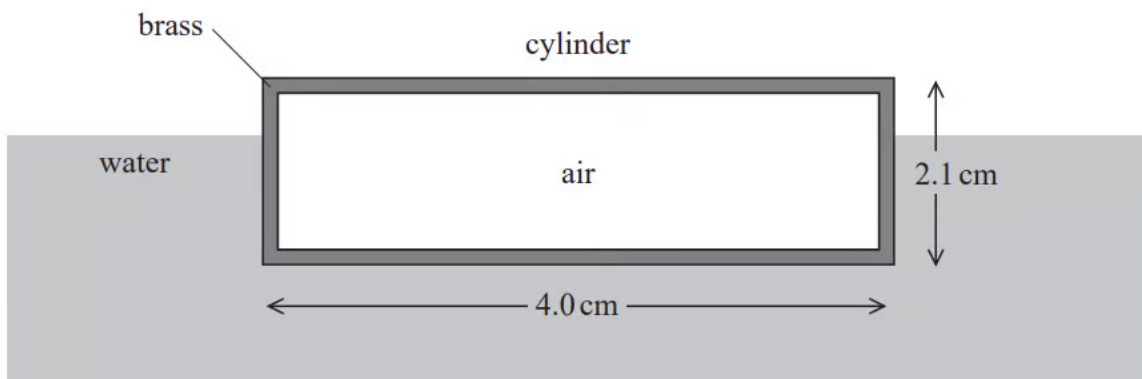


Fig. 1.1

63% of the volume of the cylinder is submerged. The cylinder contains negligible weight of air.

Explain why the brass cylinder floats.

[2 marks]

Question 5b

The density of water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

Show that the mass of the cylinder is about $9 \times 10^{-3} \text{ kg}$.

[4 marks]**Question 5c**

Deduce whether an identical hollow cylinder made of gold would also float.

Assume that the volume of gold is the same as the volume of brass.

- Density of gold = $19.3 \times 10^3 \text{ kg m}^{-3}$
- Density of brass = $8.7 \times 10^3 \text{ kg m}^{-3}$

[4 marks]



Model Answers: Medium

1a

(a)

(i) Calculate the density of the wood:

- Cross-sectional area of rod, $A = \frac{\pi d^2}{4} = \frac{\pi \times (1.5 \times 10^{-2})^2}{4} = 1.77 \times 10^{-4} \text{ m}^2$
- Volume of the rod, $V = A \times L = (1.77 \times 10^{-4}) \times (7.5 \times 10^{-2}) = 1.33 \times 10^{-5} \text{ m}^3$
[1 mark]
- Density, $\rho = \frac{m}{V} = \frac{1.3 \times 10^{-2}}{1.33 \times 10^{-5}} = 977 = 980 \text{ kgm}^{-3} \text{ (2 s.f.)}$ [1 mark]

(ii) Suggest why the designer chose this type of wood to use in their investigation:

- Value is close to the density of water; [1 mark]

Any **one** justification:

- Therefore probably similar to density of the human body; [1 mark]
- Humans are made mostly from water; [1 mark]

(iii) Suggest one change to the experimental set-up which the designer needs to make to model the situation they are investigating:

- Add salt / sugar / minerals to the water; [1 mark]
- To mimic / copy the density of sea water; [1 mark]

[Total: 6 marks]

1b

(b) Calculate the speed of the rod as its base reaches the water:

List the known quantities:

- Height, $h = 0.20 \text{ m}$
- Initial velocity, $u = 0 \text{ ms}^{-1}$
- Acceleration, $a = 9.81 \text{ ms}^{-2}$

Choose the best SUVAT equation and calculate:

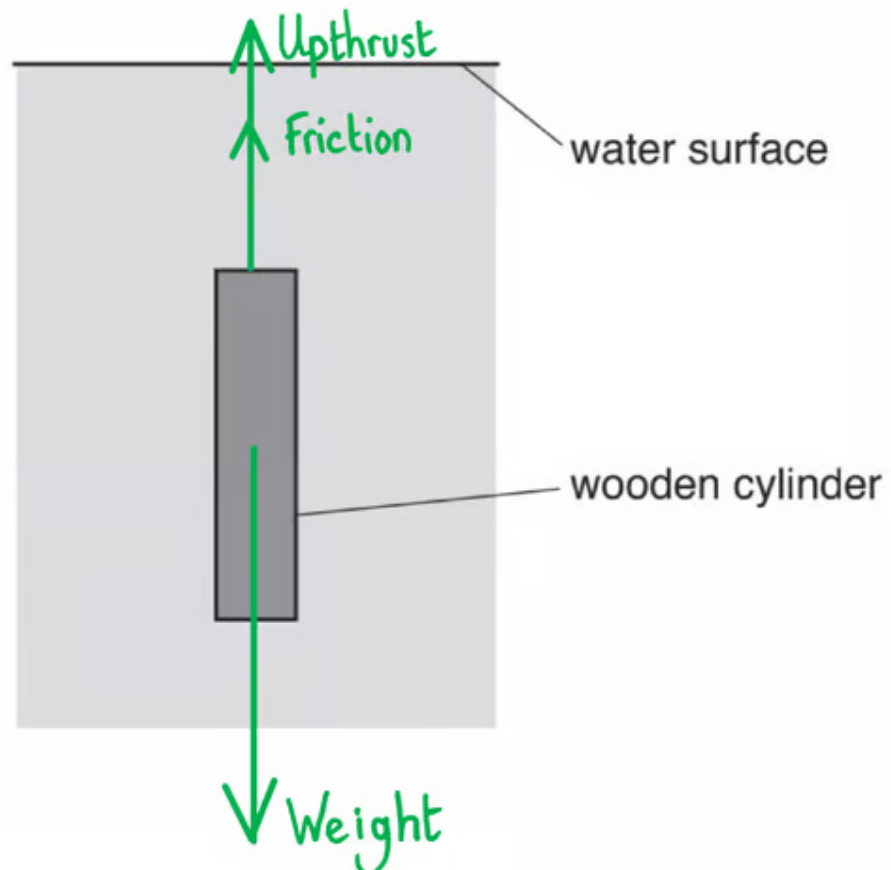
- $v^2 = 2as + u^2$
- $v = \sqrt{(2 \times 9.81 \times 0.2)}$ [1 mark]
- $v = 2.0 \text{ ms}^{-1}$ (2 s.f) [1 mark]

[Total: 2 marks]

1c

(c)

(i) The forces acting on the wooden rod should look like:



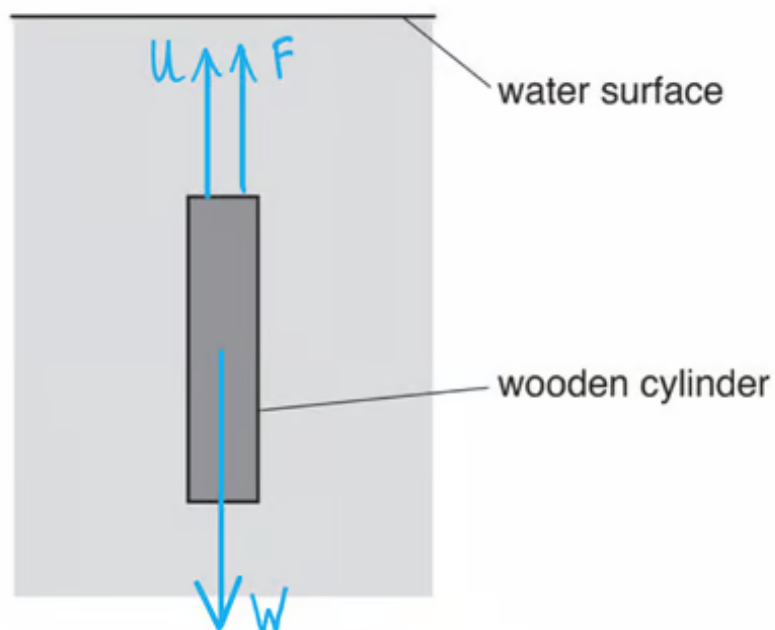
- Downwards arrow labelled W / weight / mg ; [1 mark]
- Upwards arrow labelled U / upthrust; [1 mark]
- Upwards arrow labelled D / drag / friction; [1 mark]

(ii) The resultant force on the wooden cylinder varies by:

- Resultant force decreases (with time or as the cylinder descends); [1 mark]
- Upthrust remains constant / drag decreases (as speed decreases) / resultant force is upwards / at the lowest point, drag is zero; [1 mark]
- At the deepest point, the resultant force is upwards; [1 mark]

[Total: 6 marks]

- You might have phrased the last two points in part (ii) as one sentence, saying something like 'the resultant force is always directed upwards'. This would be awarded two marks since it includes two ideas from the list
- For part (i), you may also have annotated the diagram differently. The two upwards arrows can be drawn side by side, although showing the point which they act from is a better choice



1d

(d) A likely range of pressure changes they will feel in their ears are:

Choose a range of likely heights for the guests:

- Height of adult guests, h_g : $1.5 \text{ m} \leq h \leq 2.1 \text{ m}$ (answers in range from 1.4 m to 2.2 m) [1 mark]

Calculate minimum and maximum pressures:

- Depth for calculating pressure, h : $2.5 \text{ m} \leq h \leq 1.9 \text{ m}$

Calculate pressure in a liquid, P :

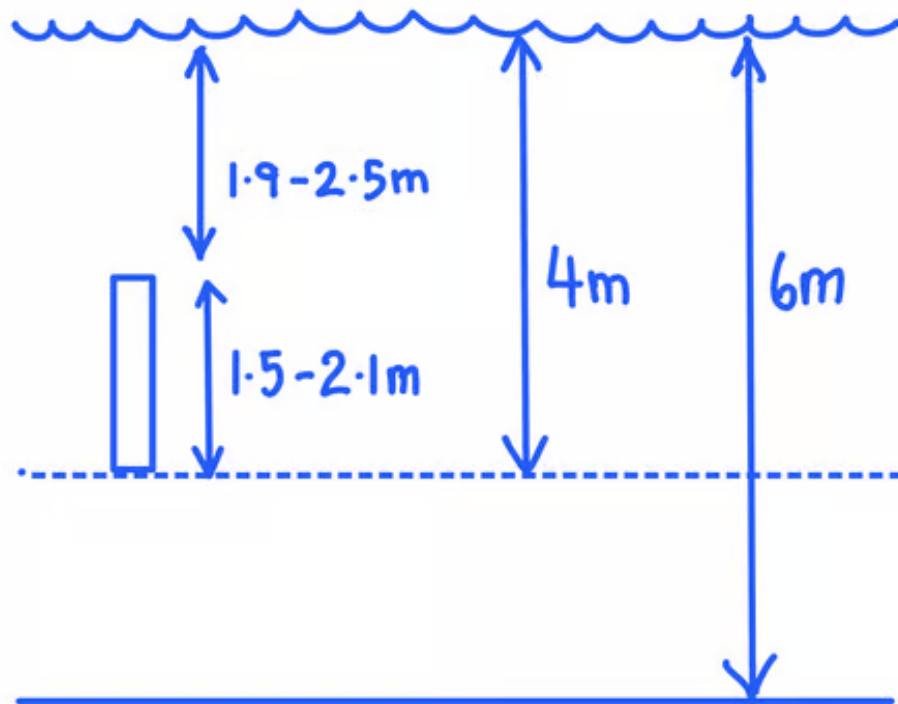
- $P = \rho g h$
- $P_{\max} = 1.03 \times 9.81 \times 2.5$ [1 mark]
- $P_{\min} = 1.03 \times 9.81 \times 1.9$ [1 mark]

State the range of pressures:

- 19.2 Pa – 25.3 Pa [1 mark]

[Total: 4 marks]

A quick sketch would help you think about the range of depths. Guests will jump in **feet** first, to a depth of **4 m**, and the question is about the pressure change in their **ears**.



We have already assumed a sensible range of heights for adults. One final assumption is to say that the ears are at this height. You could deduct 5 cm, say, from each height, but the point of assumptions and modelling is to get to **reasonable** answers more quickly and easily than by doing a full and complex calculation.

2a

(a) By modelling the balloon as a point mass, sketch a free body force diagram for the balloon and ribbon:



- Weight and tension / W and T downwards **AND** from a point, **not** a balloon; [1 mark]
- Upthrust / U upwards; [1 mark]

[Total: 2 marks]

2b

(b) Calculate the upthrust on the balloon:

Determine the volume of the balloon:

- Assume the balloon is a sphere with diameter, $r = 40 \text{ cm} = 0.40 \text{ m}$
- Volume of the balloon, $V_b = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \times \left(\frac{0.4}{2}\right)^3 = 0.0335 \text{ m}^3$ [1 mark]

Determine the upthrust, F :

- $F = \rho g V = 1.30 \times 9.81 \times 0.0335$ [1 mark]
- $F = 0.43 \text{ N}$ (2 s.f.) [1 mark]

[Total: 3 marks]

Determine the upthrust, F :

- $F = \rho g V = 1.30 \times 9.81 \times 0.0335$ [1 mark]
- $F = 0.43 \text{ N}$ (2 s.f.) [1 mark]

[Total: 3 marks]

It is reasonable to assume that the volume of the balloon can be found by treating it as a **sphere** with the same diameter.

2c

(c) A balloon filled with helium floats because:

- Helium is less dense than air; [1 mark]
- (Therefore) the weight of the displaced air is greater than the weight of the helium; [1 mark]
- Upthrust is equal to weight of displaced air; [1 mark]
- Upthrust / upwards force on the (helium-filled) balloon is larger than the weight of the balloon; [1 mark]

[Total: 4 marks]

3a

(a)

(i) The property of the liquid that changes with temperature is:

- Density; [1 mark]

(ii) Calculate the density of the liquid:

List the known quantities:

- Radius, $r = 1.2 \text{ cm} = 1.2 \times 10^{-2} \text{ m}$
- Mass of liquid displaced, $m = 7.3 \text{ g} = 7.3 \times 10^{-3} \text{ kg}$

Determine the volume of the sphere

- Volume of a sphere: $V = \frac{4}{3}\pi r^3$
- $V = \frac{4}{3}\pi(1.2 \times 10^{-2})^3 = 7.24 \times 10^{-6} \text{ m}^3$ [1 mark]

Determine the density of the liquid:

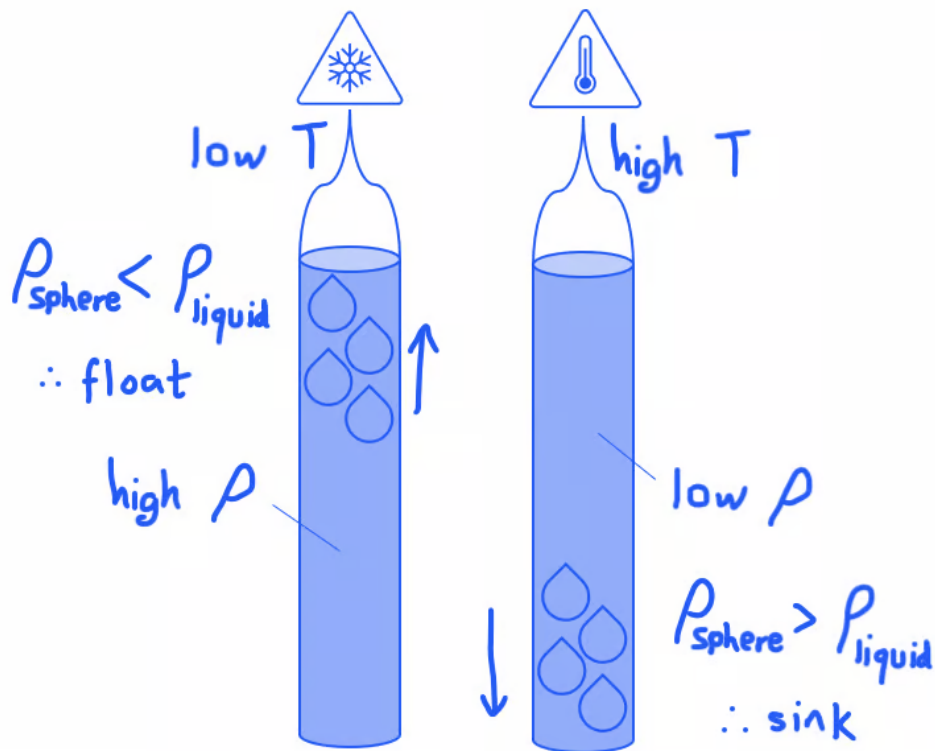
- Density: $\rho = \frac{m}{V}$
- $\rho = \frac{7.3 \times 10^{-3}}{7.24 \times 10^{-6}}$ [1 mark]
- $\rho = 1008.5 = 1010 \text{ kg m}^{-3}$ [1 mark]

[Total: 4 marks]

The basic principle of a Galileo thermometer is that the liquid in the column will become denser if the room temperature drops and less dense if the room temperature rises.

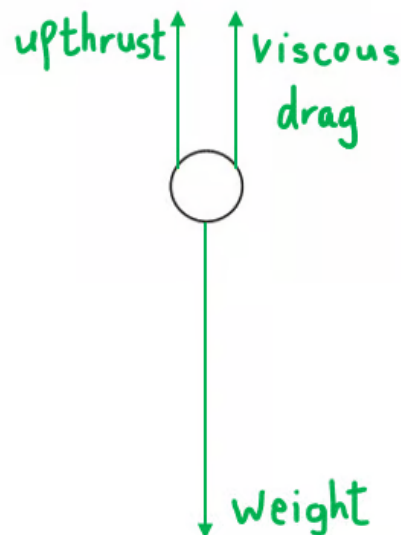
As a result, the spheres, which have a fixed density, will float or sink depending on the temperature of the room:

- An object will **float** if it is **less dense** than the fluid i.e. the spheres will float when the **temperature is low**
- An object will **sink** if it is **denser** than the fluid it is placed in i.e. the spheres will sink when the **temperature is high**



3b

(b) Arrows to show the relative size of the forces acting on the sphere as it falls should look like:



- Arrow drawn downwards **AND** labelled weight / W / mg ; [1 mark]
- Arrow drawn upwards **AND** labelled upthrust / U ; [1 mark]
- Arrow drawn upwards **AND** labelled (viscous) drag / D ; [1 mark]

- Weight is drawn visibly longer than upthrust and (viscous) drag; [1 mark]

[Total: 4 marks]

Make sure not to be sloppy with any sketches or drawings in A-level physics, as you can lose some easy marks! Remember, these forces are **vectors**, so the lengths of the arrows represent their size. Since the sphere is falling, its weight will be larger than the sum of the upwards forces (unless you're told it is at terminal velocity, in which case, the size of the arrows upwards = the size of the arrows downwards).

3c

(c) Show that the upthrust on the sphere is about 0.07 N:

List the known quantities:

- Mass of liquid displaced, $m = 7.3 \text{ g} = 7.3 \times 10^{-3} \text{ kg}$
- Acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$

Calculate the upthrust:

- Upthrust = Weight of liquid displaced; [1 mark]
- Upthrust = $mg = (7.3 \times 10^{-3}) \times 9.81$
- Upthrust = $7.16 \times 10^{-2} = 0.072 = 0.07 \text{ N}$ [1 mark]

[Total: 2 marks]

3d

(d) A sphere with a weight 0.069 N will:

- Have a net upwards force as the upthrust / U is greater than the weight / W ; [1 mark]
- (Hence) the sphere would accelerate upwards / higher **OR** stay at the top; [1 mark]

[Total: 2 marks]

This value should be compared to the weight calculated in part (c). Since it is **lower** than 0.07 N, the upthrust will be slightly higher on this sphere than one of weight 0.07 N. Always keep tabs on the values asked for in previous subparts.

4a

(a) Calculate the upthrust exerted on the submarine by the seawater:

List the known quantities:

- Volume of submarine, $V = 5.83 \times 10^3 \text{ m}^3$
- Density of seawater, $\rho = 1.03 \times 10^3 \text{ kg m}^{-3}$
- Acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$

Use Archimedes' principle equation to calculate upthrust:

- Upthrust: $U = \rho g V$ [1 mark]
- Upthrust: $U = (1.03 \times 10^3) \times 9.81 \times (5.83 \times 10^3)$
- Upthrust: $U = 5.89 \times 10^7 \text{ N}$ [1 mark]

[Total: 2 marks]

4b

(b) The mass of the submarine must be $6.0 \times 10^6 \text{ kg}$ because:

- Weight (of the submarine) is equal to the upthrust; [1 mark]

Use a calculation to justify a mass of $6.0 \times 10^6 \text{ kg}$:

EITHER

- Using weight: $W = mg \Rightarrow m = \frac{W}{g}$
- Mass: $m = \frac{5.89 \times 10^7}{9.81} = 6.0 \times 10^6 \text{ kg}$ [1 mark]

OR

- Using density: $\rho = \frac{m}{V} \Rightarrow m = \rho V$
- $m = (1.03 \times 10^3) \times (5.83 \times 10^3) = 6.0 \times 10^6 \text{ kg}$ [1 mark]

[Total: 2 marks]

4c

(c) As the submarine moves into a region of lower-density seawater:

- The upthrust (of the water on the submarine) will be less than the weight (of the submarine); [1 mark]
- Hence, a resultant force will act downwards on the submarine; [1 mark]
- Therefore, the submarine will (begin to) sink; [1 mark]

[Total: 3 marks]

4d

(d) Calculate the required change in mass of water needed to maintain its depth:

List the known quantities:

- Density of new water, $\rho = 1.01 \times 10^3 \text{ kg m}^{-3}$
- Volume of submarine, $V = 5.83 \times 10^3 \text{ m}^3$
- Initial upthrust = $5.89 \times 10^7 \text{ N}$
- Acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$

Calculate the new upthrust:

- Upthrust: $U = \rho g V$
- Upthrust: $U = (1.01 \times 10^3) \times 9.81 \times (5.83 \times 10^3) = 5.78 \times 10^7 \text{ N}$ [1 mark]

Use the net downward force to calculate the required mass to be lost:

- Resultant force: $F = (5.89 \times 10^7) - (5.78 \times 10^7) = 1.14 \times 10^6 \text{ N}$
- Required change in mass: $m = \frac{W}{g} = \frac{1.14 \times 10^6}{9.81} = 1.17 \times 10^5 \text{ kg}$ [1 mark]

[Total: 2 marks]

5a

The brass cylinder floats because:

EITHER

4d

(d) Calculate the required change in mass of water needed to maintain its depth:

List the known quantities:

- Density of new water, $\rho = 1.01 \times 10^3 \text{ kg m}^{-3}$
- Volume of submarine, $V = 5.83 \times 10^3 \text{ m}^3$
- Initial upthrust = $5.89 \times 10^7 \text{ N}$
- Acceleration of free fall, $g = 9.81 \text{ m s}^{-2}$

Calculate the new upthrust:

- Upthrust: $U = \rho g V$
- Upthrust: $U = (1.01 \times 10^3) \times 9.81 \times (5.83 \times 10^3) = 5.78 \times 10^7 \text{ N}$ [1 mark]

Use the net downward force to calculate the required mass to be lost:

- Resultant force: $F = (5.89 \times 10^7) - (5.78 \times 10^7) = 1.14 \times 10^6 \text{ N}$
- Required change in mass: $m = \frac{W}{g} = \frac{1.14 \times 10^6}{9.81} = 1.17 \times 10^5 \text{ kg}$ [1 mark]

[Total: 2 marks]

5a

The brass cylinder floats because:

EITHER

- There is an upthrust which is equal to the weight of water displaced; [1 mark]
- The upthrust is equal to the weight of the cylinder (when it's partially submerged); [1 mark]

OR

- The (overall) density (of the cylinder) is less than the density of the water; [1 mark]
- The weight of water displaced is equal to the weight of the cylinder; [1 mark]

[Total: 2 marks]

5b

(b) Show that the mass of the cylinder is about $9 \times 10^{-3} \text{ kg}$:

List the known quantities:

- Diameter of cylinder = $2.1 \text{ cm} = 2.1 \times 10^{-2} \text{ m}$
- Length of cylinder, $l = 4.0 \text{ cm} = 4.0 \times 10^{-2} \text{ m}$
- Percentage of cylinder submerged = 63%
- Density of water = $1.0 \times 10^3 \text{ kg m}^{-3}$

Determine the volume of the brass cylinder that is submerged:

- Volume of brass cylinder: $V = \pi r^2 l$
- $V = \pi \times \left(\frac{2.1 \times 10^{-2}}{2} \right)^2 \times (4.0 \times 10^{-2}) = 1.39 \times 10^{-5} \text{ m}^3$ [1 mark]
- Volume that is submerged: $V_{\text{sub}} = 0.63 \times (1.39 \times 10^{-5}) = 8.76 \times 10^{-6} \text{ m}^3$ [1 mark]

Calculate the mass of the cylinder:

- Mass of cylinder = mass of water displaced
- $\rho = \frac{m}{V} \Rightarrow m = \rho V$
- Mass of water displaced: $m = (1 \times 10^3) \times (8.76 \times 10^{-6})$ [1 mark]
- Mass of cylinder: $m = 8.76 \times 10^{-3} \text{ kg}$ [1 mark]

[Total: 4 marks]

5c

(c) Deduce whether an identical hollow cylinder made of gold would also float:

List the known quantities:

- Density of gold = $19.3 \times 10^3 \text{ kg m}^{-3}$
- Density of brass = $8.7 \times 10^3 \text{ kg m}^{-3}$

Determine the equivalent volume of gold:

- Volume of gold = volume of brass
- Volume of brass that floats: $V = \frac{m}{\rho} = \frac{8.76 \times 10^{-3}}{8.7 \times 10^3}$
- Equivalent volume of gold: $V = 1.01 \times 10^{-6} \text{ m}^3$ [1 mark]

Calculate the volume of water required to float the gold cylinder:

- Mass of gold cylinder: $m = \rho V$
- Mass of gold cylinder: $m = (19.3 \times 10^3) \times (1.01 \times 10^{-6}) = 0.0194 \text{ kg}$ [1 mark]
- Minimum volume of water required to float cylinder: $V = \frac{m}{\rho}$
- $V = \frac{0.0194}{1.00 \times 10^3} = 1.94 \times 10^{-5} \text{ m}^3$ [1 mark]

State a conclusion backed up by calculated values:

Minimum volume of water required to float cylinder = $1.94 \times 10^{-5} \text{ m}^3$

Volume of gold cylinder = $1.39 \times 10^{-5} \text{ m}^3$

- $1.94 \times 10^{-5} \text{ m}^3 > 1.39 \times 10^{-5} \text{ m}^3$, therefore the gold cylinder would sink; [1 mark]

OR

Maximum mass (or weight) of water that could be displaced = 0.014 kg (0.14 N)

Mass of the gold cylinder (or weight) = 0.019 kg (0.19 N)

- 0.019 kg > 0.014 kg **or** 0.19 N > 0.14 N, therefore the gold cylinder would sink; [1 mark]

[Total: 4 marks]